

Aim 1 - Top 10 genes by brain-met prevalence (cohort=genie; brain-met n=393, no-brain-met n=841); Fisher exact, Holm and BH-FDR across 10 tests

Gene	Mut in brain-met (n/N)	% brain-met	Mut in no-brain-met (n/N)	% no-brain-met	Prev. ratio (95% CI)	Fisher p	Holm p	BH q
TP53	206/393	52.4%	364/841	43.3%	1.21 (1.07-1.37)	0.0032	0.0324	0.0324
PIK3CA	112/393	28.5%	241/841	28.7%	0.99 (0.82-1.20)	1.0000	1.0000	1.0000
GATA3	46/393	11.7%	128/841	15.2%	0.77 (0.56-1.05)	0.1138	1.0000	0.5692
CDH1	35/393	8.9%	87/841	10.3%	0.86 (0.59-1.25)	0.4743	1.0000	0.6825
PTEN	26/393	6.6%	47/841	5.6%	1.18 (0.74-1.88)	0.5175	1.0000	0.6825
NF1	25/393	6.4%	38/841	4.5%	1.41 (0.86-2.30)	0.2109	1.0000	0.6825
ARID1A	24/393	6.1%	60/841	7.1%	0.86 (0.54-1.35)	0.5460	1.0000	0.6825
KMT2D	24/393	6.1%	54/841	6.4%	0.95 (0.60-1.52)	0.9004	1.0000	1.0000
BRCA2	24/393	6.1%	42/841	5.0%	1.22 (0.75-1.99)	0.4177	1.0000	0.6825
KMT2C	23/393	5.9%	38/841	4.5%	1.30 (0.78-2.14)	0.3254	1.0000	0.6825